$$Q_n = \frac{1}{n}[R_n - Q_n] \qquad Q_{n+1} \doteq Q_n + \alpha[R_n - Q_n] \qquad Q_{n+1} = (1-\alpha)^n Q_1 + \sum_{i=1}^{n} \alpha(1-\alpha)^{n-i} R_i$$

$$A_t \doteq \arg\max_a \left[Q_t(a) + c\sqrt{\frac{\ln t}{N_t(a)}} \right] \qquad p(s', r|s, a) \doteq \Pr\{S_t = s', R_t = r|S_{t-1} = s, A_{t-1} = a\}$$

$$p(s'|s, a) \doteq \Pr\{S_t = s'|S_{t-1} = s, A_{t-1} = a\} = \sum_{r\in\mathcal{R}} p(s', r|s, a) \qquad v_\pi(s) = \sum_a \pi(a|s) q_\pi(s, a)$$

$$r(s, a) \doteq \mathbb{E}[R_t|S_{t-1} = s, A_{t-1} = a] = \sum_{r\in\mathcal{R}} r \sum_{s'\in\mathcal{S}} p(s', r|s, a) \qquad q_\pi(s, a) = \sum_{s',r} p(s', r|s, a)\Big[r + \gamma v_\pi(s')\Big]$$

$$r(s, a, s') \doteq \mathbb{E}[R_t|S_{t-1} = s, A_{t-1} = a, S_t = s'] = \sum_{r\in\mathcal{R}} r \frac{p(s', r|s, a)}{p(s'|s, a)} \qquad q_*(s, a) = \sum_{s',r} p(s', r|s, a)\Big[r + \gamma \max_{a'} q_*(s', a')\Big]$$

$$G_t \doteq \sum_{k=t+1}^{T} \gamma^{k-t-1} R_k \qquad G_t = R_{t+1} + \gamma G_{t+1} \qquad q_*(s, a) = \max_\pi q_\pi(s, a) = \mathbb{E}[R_{t+1} + \gamma v_*(S_{t+1})|S_t = s, A_t = a]$$

$$v_\pi(s) \doteq \mathbb{E}_\pi[G_t|S_t = s] = \sum_a \pi(a|s) \sum_{s',r} p(s', r|s, a)\Big[r + \gamma v_\pi(s')\Big] \qquad v_*(s) = \max_a \Big[r(s, a) + \gamma \sum_{s'} p(s'|s, a) v_*(s')\Big]$$

$$q_\pi(s, a) \doteq \mathbb{E}_\pi[G_t|S_t = s, A_t = a] = \mathbb{E}_\pi\left[\sum_{k=0}^{\infty} \gamma^k R_{t+k+1} \Big| S_t = s, A_t = a\right] = \sum_{s',r} p(s', r)\Big[r + \gamma \sum_{a'} \pi(a'|s') q_\pi(s', a')\Big]$$

$$v_*(s) = \max_{a\in\mathcal{A}(s)} q_{\pi_*}(s, a) = \max_a \sum_{s',r} p(s', r|s, a)[r + \gamma v_*(s')] \qquad q_*(s, a) = \sum_{s,r} p(s', r|s, a)\Big[r + \gamma v_*(s')\Big]$$

$$\pi_*(s) = \arg\max_a q_*(s, a) = \arg\max_a \sum_{s',r} p(s', r|s, a)\Big[r + \gamma v_*(s')\Big] \qquad q_*(s, a) = r(s, a) + \gamma \sum_{s'} p(s'|s, a) \max_{a'} q_*(s', a')$$

$$q(s, a) = r(s, a) + \gamma \sum_{s'} p(s'|s, a) \sum_{a'} \pi(a'|s') q_\pi(s', a') \qquad v_\pi(s) = \sum_a \pi(a|s)\Big[r(s, a) + \gamma \sum_{s'} p(s'|s, a) v_\pi(s')\Big]$$

$$V(S_t) \leftarrow V(S_t) + \alpha\Big[G_t - V(S_t)\Big] \qquad \rho_{t:T-1} \doteq \prod_{k=t}^{T-1} \frac{\pi(A_k|S_k)}{b(A_k|S_k)} \qquad \mathbb{E}[\rho_{t:T-1} G_t|S_t = s] = v_\pi(s)$$

$$V(s) \doteq \frac{\sum_{t\in\mathcal{T}(s)} \rho_{t:T(t)-1} G_t}{|\mathcal{T}(s)|} \qquad V(s) \doteq \frac{\sum_{t\in\mathcal{T}(s)} \rho_{t:T(t)-1} G_t}{\sum_{t\in\mathcal{T}(s)} \rho_{t:T(t)-1}} \qquad G_t - V(S_t) = \sum_{k=t}^{T-1} \gamma^{k-t} \delta_k$$

$$V(S_t) \leftarrow V(S_t) + \alpha\Big[R_{t+1} + \gamma V(S_{t+1}) - V(S_t)\Big] \qquad \delta_t \doteq R_{t+1} + \gamma V(S_{t+1}) - V(S_t)$$

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha\Big[R_{t+1} + \gamma Q(S_{t+1}, A_{t+1}) - Q(S_t, A_t)\Big]$$

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha\Big[R_{t+1} + \gamma \max_a Q(S_{t+1}, a) - Q(S_t, A_t)\Big]$$

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha\Big[R_{t+1} + \gamma \sum_a \pi(a|S_{t+1}) Q(S_{t+1}, a) - Q(S_t, A_t)\Big]$$

$$Q_1(S_t, A_t) \leftarrow Q_1(S_t, A_t) + \alpha\Big[R_{t+1} + \gamma Q_2(S_{t+1}, \arg\max_a Q_1(S_{t+1}, a)) - Q_1(S_t, A_t)\Big]$$

$$\overline{VE}(\mathbf{w}) \doteq \sum_{s\in\mathcal{S}} \mu(s)\Big[v_\pi(s) - \hat{v}(s,\mathbf{w})\Big]^2 \qquad \hat{v}(s,\mathbf{w}) \doteq \mathbf{w}^\top \mathbf{x}(s) \qquad \nabla \mathbf{w}^\top \mathbf{x}(s) = \mathbf{x}(s)$$

$$\eta(s) = h(s) + \sum_{\bar{s}} \eta(\bar{s}) \sum_a \pi(a|\bar{s}) p(s|\bar{s},a) \qquad \mu(s) = \frac{\eta(s)}{\sum_{s'} \eta(s')} \qquad \sum_s \mu_\pi(s) \sum_a \pi(a|s) p(s'|s,a) = \mu_\pi(s')$$

$$\mathbf{w}_{t+1} \doteq \mathbf{w}_t + \alpha\big[U_t - \hat{v}(S_t,\mathbf{w}_t)\big]\nabla \hat{v}(S_t,\mathbf{w}_t) \qquad r(\pi) = \sum_s \mu_\pi(s) \sum_a \pi(a|s) \sum_{s',r} p(s',r|s,a) r$$

$$\mathbf{w}_{t+1} \doteq \mathbf{w}_t + \alpha\Big(R_{t+1} + \gamma \mathbf{w}_t^\top \mathbf{x}_{t+1} - \mathbf{w}_t^\top \mathbf{x}_t\Big)\mathbf{x}_t \qquad \mathbb{E}[\mathbf{w}_{t+1}|\mathbf{w}_t] = \mathbf{w}_t + \alpha(\mathbf{b} - \mathbf{A}\mathbf{w}_t)$$

$$\mathbf{w}_{t+1} \doteq \mathbf{w}_t + \alpha\Big[R_{t+1} + \gamma \hat{q}(S_{t+1},A_{t+1},\mathbf{w}_t) - \hat{q}(S_t,A_t,\mathbf{w}_t)\Big]\nabla \hat{q}(S_t,A_t,\mathbf{w}_t)$$

$$J(\boldsymbol{\theta}) \doteq v_{\pi_{\boldsymbol{\theta}}}(s_0) \qquad G_t \doteq R_{t+1} - r(\pi) + R_{t+2} - r(\pi) + R_{t+3} - r(\pi) + \ldots \qquad \Pr\{A_t = a\} \doteq \frac{e^{H_t(a)}}{\sum_{b=1}^k e^{H_t(b)}} \doteq \pi_t(a)$$

$$v_\pi(s) = \sum_a \pi(a|s) \sum_{r,s'} p(s',r|s,a)\Big[r - r(\pi) + v_\pi(s')\Big] \qquad \boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \alpha \widehat{\nabla J(\boldsymbol{\theta}_t)} \qquad \sigma(s,\boldsymbol{\theta}) \doteq \exp\Big(\boldsymbol{\theta}_\sigma^\top \mathbf{x}_\sigma(s)\Big)$$

$$q_\pi(s,a) = \sum_{r,s'} p(s',r|s,a)\Big[r - r(\pi) + \sum_{a'} \pi(a'|s') q_\pi(s',a')\Big] \qquad \boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \alpha\Big(G_t - b(S_t)\Big)\frac{\nabla \pi(A_t|S_t,\boldsymbol{\theta}_t)}{\pi(A_t|S_t,\boldsymbol{\theta}_t)}$$

$$v_*(s) = \max_a \sum_{r,s'} p(s',r|s,a)\Big[r - \max_\pi r(\pi) + v_*(s')\Big] \qquad \pi(a|s,\boldsymbol{\theta}) \doteq \frac{e^{h(s,a,\boldsymbol{\theta})}}{\sum_b e^{h(s,b,\boldsymbol{\theta})}} \qquad \mu(s,\boldsymbol{\theta}) \doteq \boldsymbol{\theta}_\mu^\top \mathbf{x}_\mu(s)$$

$$q_*(s,a) = \sum_{r,s'} p(s',r|s,a)\Big[r - \max_\pi r(\pi) + \max_{a'} q_*(s',a')\Big] \qquad \boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \alpha\gamma^t G \nabla \ln \pi(A_t|S_t,\boldsymbol{\theta})$$

$$\delta_t \doteq R_{t+1} - \overline{R}_{t+1} + \hat{v}(S_{t+1},\mathbf{w}_t) - \hat{v}(S_t,\mathbf{w}_t) \qquad \boldsymbol{\theta}_{t+1} = \boldsymbol{\theta}_t + \alpha G_t \frac{\nabla \pi(A_t|S_t,\boldsymbol{\theta}_t)}{\pi(A_t|S_t,\boldsymbol{\theta})}$$

$$\nabla J(\boldsymbol{\theta}) \propto \sum_s \mu(s) \sum_a q_\pi(s,a) \nabla \pi(a|s,\boldsymbol{\theta}) \qquad \nabla J(\boldsymbol{\theta}) = \mathbb{E}_\pi\left[G_t \frac{\nabla \pi(A_t|S_t,\boldsymbol{\theta}_t)}{\pi(A_t|S_t,\boldsymbol{\theta})}\right]$$

$$\boldsymbol{\theta}_{t+1} \doteq \boldsymbol{\theta}_t + \alpha\Big(G_{t:t+1} - \hat{v}(S_t,\boldsymbol{w})\Big)\frac{\nabla \pi(A_t|S_t,\boldsymbol{\theta}_t)}{\pi(A_t|S_t,\boldsymbol{\theta}_t)} = \boldsymbol{\theta}_t + \alpha\Big(R_{t+1} + \gamma \hat{v}(S_{t+1},\mathbf{w}) - \hat{v}(S_t,\mathbf{w})\Big)\frac{\nabla \pi(A_t|S_t,\boldsymbol{\theta}_t)}{\pi(A_t|S_t,\boldsymbol{\theta}_t)}$$

$$p(x) \doteq \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \qquad \pi(a|s,\boldsymbol{\theta}) \doteq \frac{1}{\sigma(s,\boldsymbol{\theta})\sqrt{2\pi}} \exp\left(-\frac{(a-\mu(s,\boldsymbol{\theta}))^2}{2\sigma(s,\boldsymbol{\theta})^2}\right)$$

$$H_{t+1}(A_t) \doteq H_t(A_t) + \alpha(R_t - \hat{R}_t)(1 - \pi_t(A_t)) \;\text{ and }\; H_{t+1}(a) \doteq H_t(a) - \alpha(R_t - \hat{R}_t)\pi_t(A_t) \;\; \forall a \neq A_t$$